Effects of increasing amounts of dietary cholesterol on postprandial lipemia and lipoproteins in human subjects.

Our aim was to determine the effects of increasing amounts of dietary cholesterol (0-710 mg) on the postprandial plasma lipid responses and lipoprotein changes in normolipidemic human subjects. Ten subjects were fed five different test meals in a random order: one meal did not contain fat or cholesterol while the four others contained a fixed amount of lipids (45 g) and 0, 140, 280, and 710 mg cholesterol, respectively. Fasting and post-meal blood samples were obtained for 7 h. Large and small triglyceride-rich lipoproteins (TRL), low density (LDL), and high density (HDL) lipoproteins were isolated. Compared to the no-fat, no-cholesterol meal, the fat-enriched meals raised ($P < 0.05$) plasma triglycerides, phospholipids, and free cholesterol and lowered cholesteryl esters postprandially. The meals containing zero or 140 mg cholesterol generally elicited comparable postprandial plasma and lipoprotein lipid responses. The meals providing 280 or 710 mg cholesterol significantly increased postprandial plasma phospholipids and large TRL triglycerides and decreased plasma esterified cholesterol. The lipid composition of the large TRLs and the concentrations of the small TRL lipid components were not altered postprandially by cholesterol intake. On the other hand, LDL free cholesterol increased after 3 h, LDL cholesteryl esters dropped after 3 and 7 h, HDL cholesteryl esters dropped after 3 h, and HDL phospholipids increased 7 h after ingesting meals highly enriched in cholesterol. Blood insulin, apoA-I and apoB were not altered postprandially by cholesterol intake. Thus, the data show that ingesting more than 140 mg cholesterol per meal significantly alters the postprandial lipoprotein response in healthy subjects.